3DWEBSOFT INC. FOUNDATION

FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Class 40: Python Assessment

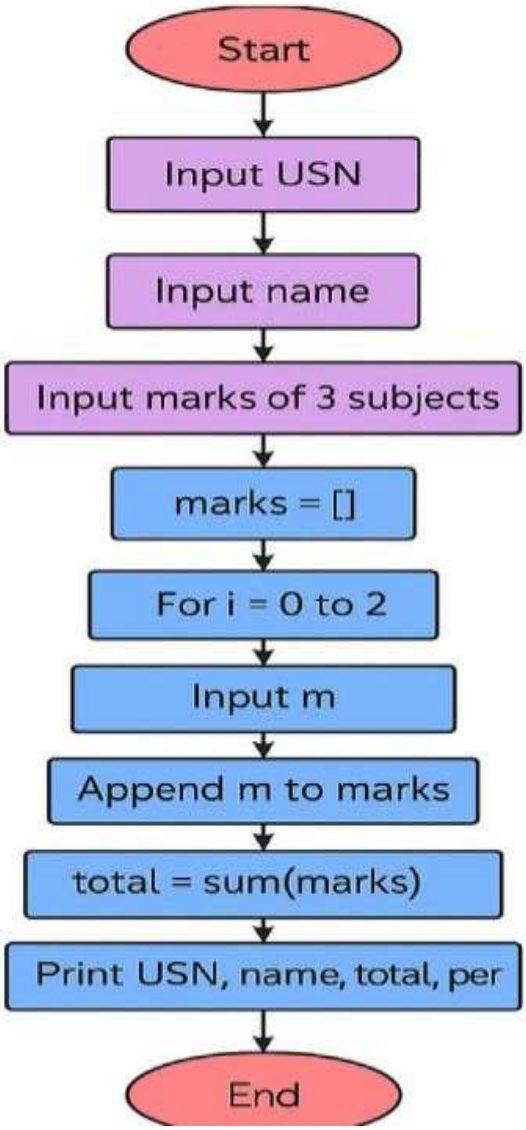


1. Assessment Objectives

This assessment checks whether students can:

- Understand basic Python concepts.
- Write simple Python programs.
- Use variables and data types.
- Take input and display output.
- Use operators.
- Apply if, elif, and else.
- Use for and while loops.
- Work with Lists and Strings.
- Create and use Functions.
- Perform basic File Handling.
- Handle simple Exceptions.
- Solve programming problems independently.





2. Topics Covered

No.	Topic	Skills
1	Python Basics	Python introduction
2	Variables	Store information

No.	Topic	Skills
3	Data Types	int, float, str, bool
4	Input & Output	input(), print()
5	Operators	Arithmetic and comparison
6	Conditions	if, elif, else
7	Loops	for, while
8	Functions	def, parameters, return
9	Lists	Store multiple values
10	Strings	String methods
11	File Handling	Read/write files
12	Exception Handling	try, except
13	Mini Projects	Problem solving

PART A – THEORY ASSESSMENT

Section 1: Multiple Choice Questions

1. What is Python?

A) A computer game
B) A programming language
C) A keyboard
D) An operating system

Answer: B

2. Which function displays output?

A) input()

B) display()

C) print()

D) show()

Answer: C

3. Which function takes input from the user?

A) input()

B) get()

C) read()

D) scan()

Answer: A

4. Which data type stores whole numbers?

A) str

B) float

C) int

D) bool

Answer: C

5. Which symbol is used for multiplication?

A) x

B) *

C) #

D) %

Answer: B

6. Which operator gives the remainder?

A) /

B) //

C) %

D) **

Answer: C

7. Which keyword is used for a condition?

A) if

B) check

C) when

D) condition

Answer: A

8. Which keyword is used to create a function?

A) function

B) def

C) fun

D) create

Answer: B

9. Which brackets are used for a List?

A) ( )

B) { }

C) [ ]

D) < >

Answer: C

10. What is the first List index?

A) 0

B) 1

C) -1

D) 10

Answer: A

11. Which method adds an item to the end of a List?

A) add()

B) append()

C) insert()

D) push()

Answer: B

12. Which mode is used to read a file?

A) r

B) w

C) a

D) x

Answer: A

13. Which mode is used to add information to an existing file?

A) r

B) w

C) a

D) x

Answer: C

14. Which block is used to handle exceptions?

A) try/except

B) if/else

C) for/while

D) def/return

Answer: A

15. Which exception can happen when dividing by zero?

A) ValueError

B) IndexError

C) ZeroDivisionError

D) NameError

Answer: C

Fill in the Blanks

1.

Python is a _____ language.

Answer: programming

2.

We use _____ to display output.

Answer: print()

3.

We use _____ to get input.

Answer: input()

4.

The first index of a List is _____.

Answer: 0

5.

The keyword used to create a function is _____.

Answer: def

6.

The List method used to add an item is _____.

Answer: append()

7.

The file mode used for reading is _____.

Answer: r

8.

The file mode used for writing is _____.

Answer: w

9.

The keyword used to handle an exception is _____.

Answer: except

10.

Finding and fixing programming errors is called _____.

Answer: debugging

True or False

Statement	Answer
Python is a programming language.	✅ True
print() takes input from the user.	❌ False
List indexing starts at 0.	✅ True
append() adds an item to a List.	✅ True
w means Read mode.	❌ False
r means Read mode.	✅ True
Functions are created using def.	✅ True
try and except are used for exception handling.	✅ True
Python ignores indentation.	❌ False
A List can contain multiple values.	✅ True

Short Answer Questions

Q1. What is a variable?

A **variable** is a name used to store information.

name = "Aisha"

age = 13

Q2. What is a List?

A List is a collection of multiple items.

fruits = ["Apple", "Mango", "Banana"]

Q3. What is a function?

A function is a reusable block of code that performs a particular task.

```
def greet():
    print("Hello!")
```

Q4. What is a loop?

A loop repeats a block of code.

Example:

```
for i in range(5):
    print(i)
```

Q5. What is exception handling?

Exception handling allows a program to safely handle unexpected errors.

```
try:
    number = int(input("Enter number: "))
except ValueError:
    print("Invalid number")
```

PART B – OUTPUT PREDICTION

Question 1

What is the output?

```
x = 10
y = 5
```

```
print(x + y)
```

Answer:

15

Question 2

```
name = "Python"

print(name)
```

Answer:

Python

Question 3

```
numbers = [10, 20, 30]

print(numbers[1])
```

Answer:

20

Question 4

```
for i in range(1, 4):
    print(i)
```

Answer:

1

2

3

Question 5

```
x = 10

if x > 5:
    print("Big")
else:
    print("Small")
```

Answer:

Big

PART C – FIND THE ERROR

Question 1

```
print("Hello"
```

Error:

Missing closing).

Correct:

```
print("Hello")
```

Question 2

```
if age >= 10
    print("Child")
```

Error:

Missing :.

Correct:

```
if age >= 10:
    print("Child")
```

Question 3

```
for i in range(5)
    print(i)
```

Error:

Missing :.

Correct:

```
for i in range(5):
    print(i)
```

PART D – PRACTICAL ASSESSMENT

Students should complete these programs on the computer.

Practical 1 – Greeting Program

Write a program that asks the student's name and displays:

Hello Aisha!

Welcome to Python!

Expected Code

```
name = input("Enter your name: ")


print("Hello", name)
print("Welcome to Python!")
```

Practical 2 – Add Two Numbers

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("Sum:", a + b)
```

Practical 3 – Even or Odd

```
number = int(input("Enter a number: "))

if number % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Practical 4 – Grade Calculator

```
marks = int(input("Enter marks: "))

if marks >= 90:
    print("Grade A+")
elif marks >= 80:
    print("Grade A")
elif marks >= 70:
    print("Grade B")
elif marks >= 60:
    print("Grade C")
```

```
elif marks >= 40:
    print("Grade D")
else:
    print("Fail")
```

Practical 5 – Multiplication Table

```
number = int(input("Enter number: "))

for i in range(1, 11):
    print(number, "×", i, "=", number * i)
```

Practical 6 – List Practice

Create a List containing five fruits and display each fruit.

```
fruits = ["Apple", "Mango", "Banana", "Orange", "Grapes"]

for fruit in fruits:
    print(fruit)
```

Practical 7 – Function

Create a function that greets a student.

```
def greet(name):
    print("Hello", name)

greet("Aisha")
```

Practical 8 – File Handling

Create a file and write a message into it.

```
with open("message.txt", "w") as file:
    file.write("I am learning Python!")

print("Message saved.")
```

Practical 9 – Exception Handling

Write a program that safely accepts a number.

```
try:
    number = int(input("Enter a number: "))
    print("You entered:", number)

except ValueError:
    print("Please enter numbers only.")
```

PART E – CODING CHALLENGES

Challenge 1 – Largest Number

Write a program to find the largest of two numbers.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

if a > b:
    print("Largest:", a)
else:
```

```
print("Largest:", b)
```

Challenge 2 – Sum of List

```
numbers = [10, 20, 30, 40, 50]

print("Total:", sum(numbers))
```

Challenge 3 – Highest Mark

```
marks = [78, 92, 85, 67, 95]

print("Highest:", max(marks))
```

Challenge 4 – Reverse a String

```
word = input("Enter a word: ")

print("Reverse:", word[::-1])
```

PART F – MINI PROJECT ASSESSMENT

Project: Student Performance Analyzer

Students must create a program that:

Input

- Student name
- Class
- Five subject marks

Process

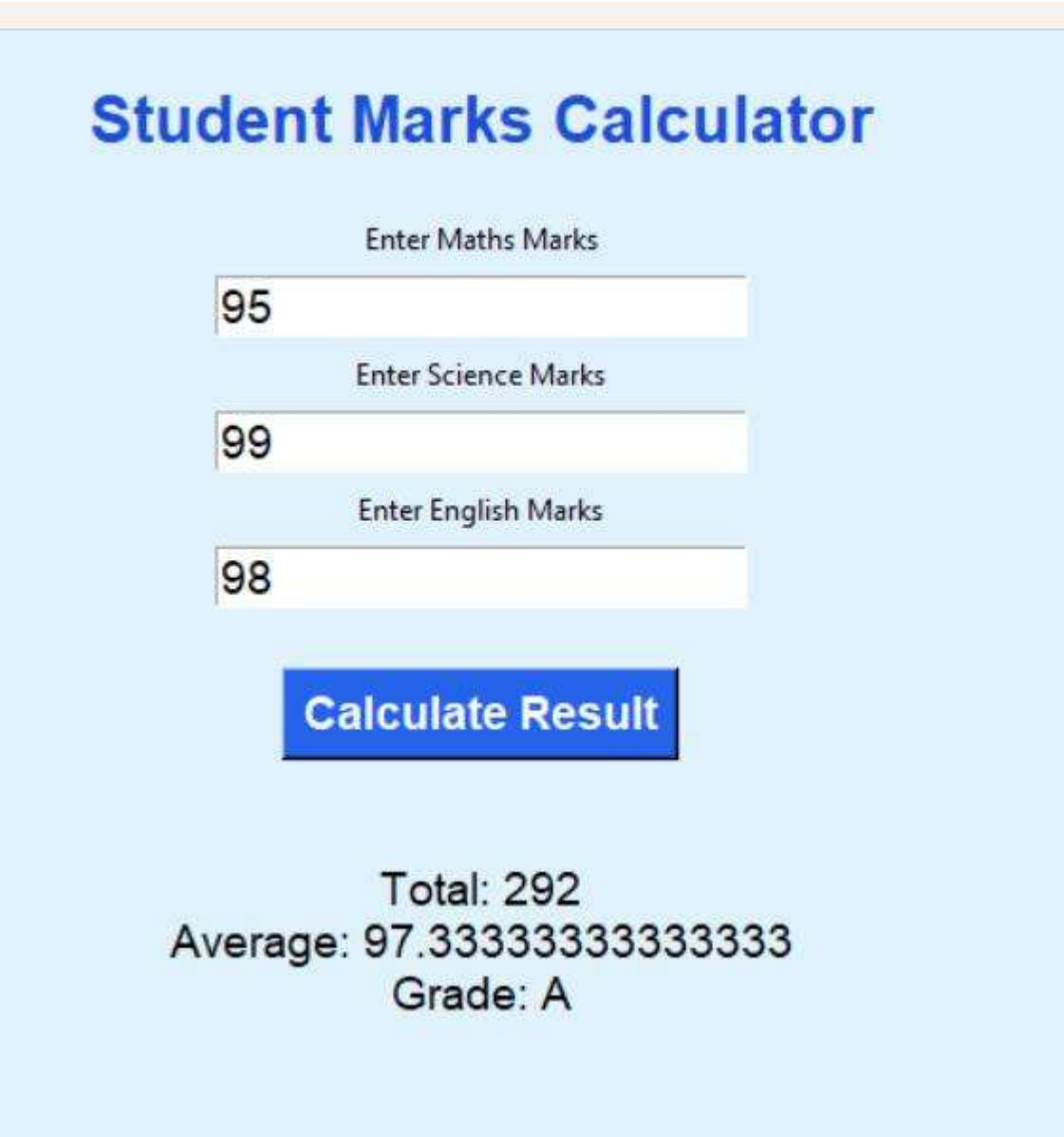
Calculate:

- Total
- Average
- Highest mark
- Lowest mark
- Grade

Output

Display a complete report.





Suggested Code

```
print("================================")
print("      STUDENT PERFORMANCE")
print("================================")


name = input("Enter student name: ")
student_class = input("Enter class: ")
```

```
marks = []

try:
    for i in range(5):
        mark = int(input("Enter marks: "))
        marks.append(mark)

    total = sum(marks)
    average = total / len(marks)

    if average >= 90:
        grade = "A+"
    elif average >= 80:
        grade = "A"
    elif average >= 70:
        grade = "B"
    elif average >= 60:
        grade = "C"
    elif average >= 40:
        grade = "D"
    else:
        grade = "F"

    print()
    print("================================")
```

```
    print("       REPORT CARD")
    print("==============================")
    print("Name:", name)
    print("Class:", student_class)
    print("Marks:", marks)
    print("Total:", total)
    print("Average:", round(average, 2))
    print("Highest:", max(marks))
    print("Lowest:", min(marks))
    print("Grade:", grade)

except ValueError:
    print("Please enter valid marks.")

finally:
    print("Assessment completed!")
```

VIVA QUESTIONS

The teacher can ask each student these questions.

Basic Questions

1. What is Python?
2. What is a variable?
3. What is a data type?
4. What does print() do?
5. What does input() do?

Conditions

6. What is an if statement?
7. What is the difference between if and else?
8. Why do we use elif?

Loops

9. What is a loop?
10. What is the difference between for and while?

Lists

11. What is a List?
12. What is the first List index?
13. What does append() do?

Functions

14. What is a function?
15. Which keyword creates a function?

Files

16. What is File Handling?
17. What does "r" mean?
18. What does "w" mean?
19. What does "a" mean?

Exceptions

20. What is an exception?
21. What does try do?
22. What does except do?

PART H – PRACTICAL TEST

Give students **one problem** from each category.

Beginner

Question: Create a program to calculate the area of a rectangle.

Area = Length × Width

Intermediate

Question: Create a program to enter five marks and display:

- Total
- Average
- Highest
- Lowest

Challenge

Question: Create a simple quiz program with five questions and calculate the student's score.

Assessment Marking Scheme

Total: 100 Marks

Section	Marks
Theory MCQs	15
Fill in the blanks	10
Short answers	10
Output prediction	10
Error finding	10

Section	Marks
Practical coding	25
Mini project	10
Viva	10
Total	**100**

Performance Levels

Marks	Level	Description
90–100	Excellent	Excellent Python understanding
80–89	Very Good	Strong programming skills
70–79	Good	Good understanding
60–69	Developing	Needs more practice
40–59	Basic	Requires additional support
Below 40	Needs Practice	More revision required

Student Assessment Sheet

Student Name: _____

Class: _____

Date: _____

Skill	Score
Python Basics	/10
Variables & Data Types	/10
Input & Output	/10

Skill	Score
Operators	/10
Conditions	/10
Loops	/10
Lists & Strings	/10
Functions	/10
File & Exception Handling	/10
Project & Viva	/10
Total	**/100**

Teacher Observation Checklist

Skill	Yes	Needs Practice
Understands Python syntax	☐	☐
Uses variables correctly	☐	☐
Uses input()	☐	☐
Uses print()	☐	☐
Writes conditions	☐	☐
Uses loops	☐	☐
Uses Lists	☐	☐
Creates functions	☐	☐
Handles errors	☐	☐
Can explain own code	☐	☐

Skill	Yes	Needs Practice
Can solve a new problem	☐	☐

Assessment Tips for Students

Before coding:

1. Read the question carefully.

2. Identify the input.

3. Identify the required output.

4. Think about the steps.

5. Start coding.

If you get an error:

Don't panic!

Read Error

↓

Find Line

↓

Check Code

↓

Fix Error

↓

Run Again

Final Coding Assessment

Build Your Own Program

Students should independently choose one:

Option 1

Number Guessing Game

Option 2

Simple Calculator

Option 3

Student Report Card

Option 4

Quiz Game

Option 5

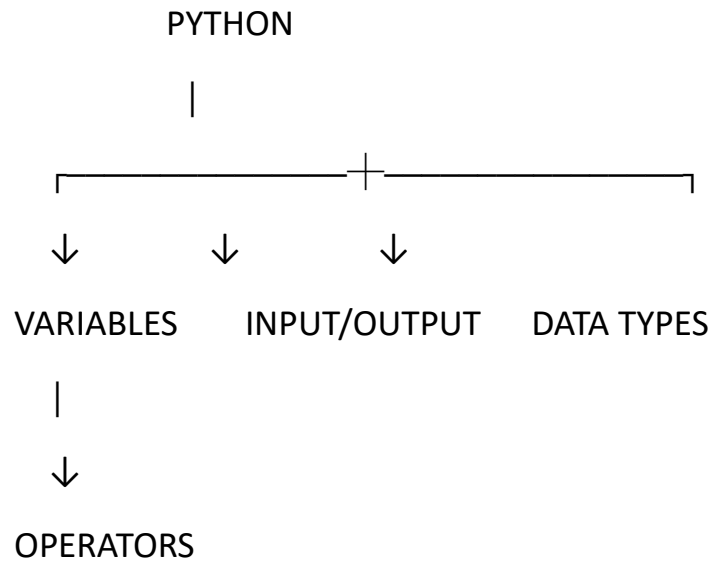
Shopping List

Option 6

To-Do List

The program should include at least **four Python concepts** learned during the course.

Final Revision Map



|

↓

CONDITIONS

|

↓

LOOPS

|

↓

LISTS & STRINGS

|

↓

FUNCTIONS

|

↓

FILE HANDLING

|

↓

EXCEPTION HANDLING

|

↓

MINI PROJECT

|

↓

ASSESSMENT